MATERNAL-NEWBORN NURSING

FETAL HEART RATE

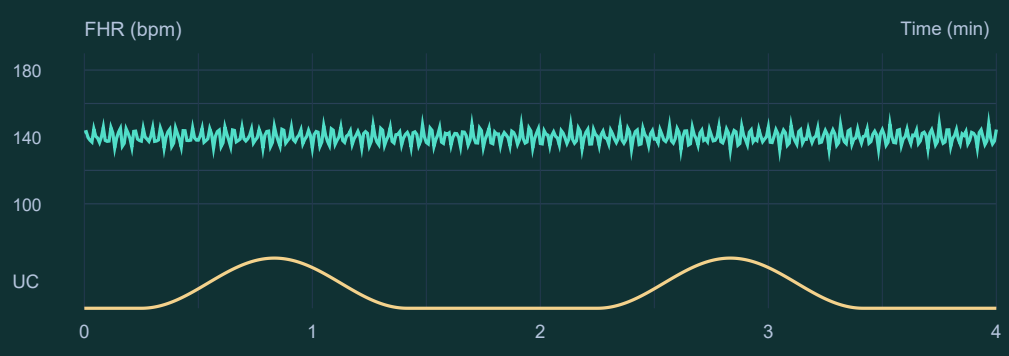
TRACING CATEGORIES / U.S. NICHD + ACOG system during labor

SCAN: baseline → variability → accelerations/decelerations → contractions → trend

I CATEGORY I

NORMAL

ALL must fit:
Baseline **110-160 bpm**
Moderate variability: 6-25 bpm
No late or variable decelerations
Early dips: present or absent
Accelerations: present or absent



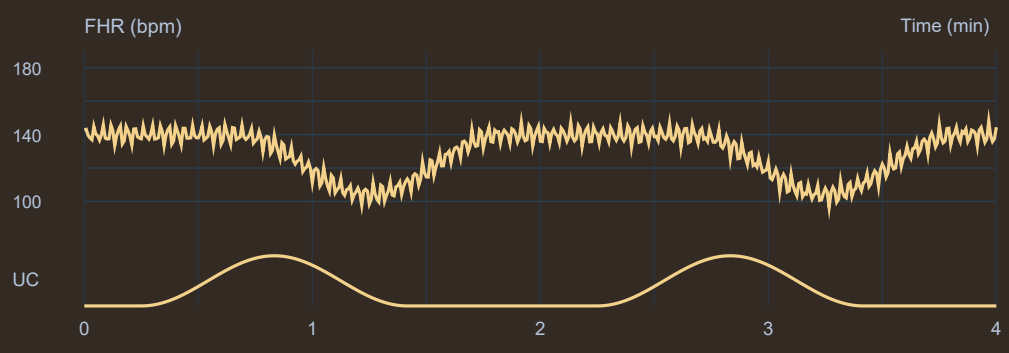
Example: baseline 140 + moderate variability; no decelerations.

MEANING / ACTION: Reassuring acid-base status now. Continue routine intrapartum care.

II CATEGORY II

INDETERMINATE

Does not fit I or III.
Examples:
Tachycardia (**>160 bpm**)
Minimal or marked variability
Recurrent late/variable dips with variability remaining
A prolonged deceleration



Example: late decelerations + moderate variability.

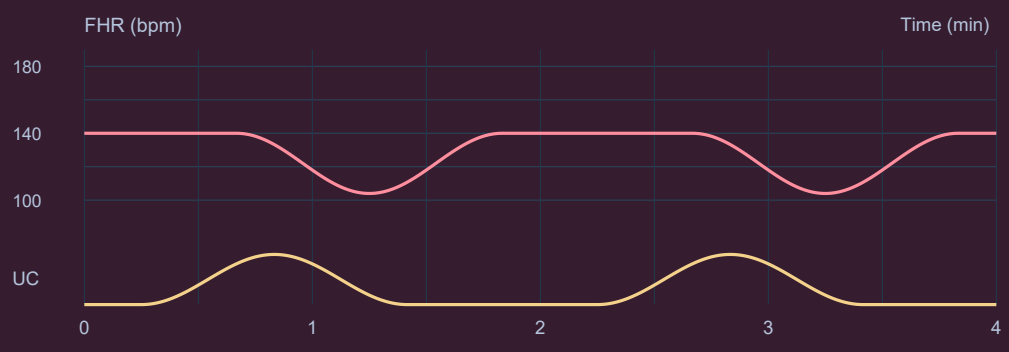
ACTION: Evaluate the cause; correct reversible problems and reassess. Category II alone does not diagnose acidemia or require cesarean birth.

III CATEGORY III

ABNORMAL

ABSENT variability + one:
- Recurrent late decelerations
- Recurrent variable decelerations
- Bradycardia (**<110 bpm**)

OR a true sinusoidal pattern



Example: late decelerations + ABSENT variability.



TRUE SINUSOIDAL: smooth, regular waves;
3-5 cycles/min for ≥20 minutes.
This is a separate route to Category III.

ACT NOW: Immediate team evaluation + corrective measures. If Category III does not resolve, prepare for expedited birth.

VARIABILITY MATTERS

Absent variability **alone ≠ III**.
Look for bradycardia or recurrent dips.

LATE ≠ ALWAYS III

With moderate variability, recurrent late decelerations are **Category II**.

REASSESS

Categories can change.
Use the full tracing and clinical picture.

Teaching excerpts: assume the shown patterns persist across the assessment window. Baseline needs a 10-minute review; recurrent dips are judged over 20 minutes. UC = contractions; height shown is relative.

Sources [1, 2]. Original schematic tracings; not patient recordings. See page 2 for patterns, responses and references.

FETAL HEART RATE

READ THE PATTERNS / then match your response to the cause

01 VARIABILITY = IRREGULAR BASELINE FLUCTUATIONS

[2, 3]

ABSENT

Undetectable

MINIMAL

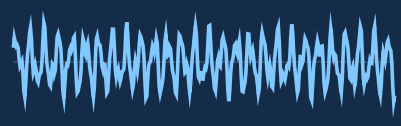
Detectable, ≤5 bpm

MODERATE



6-25 bpm

MARKED

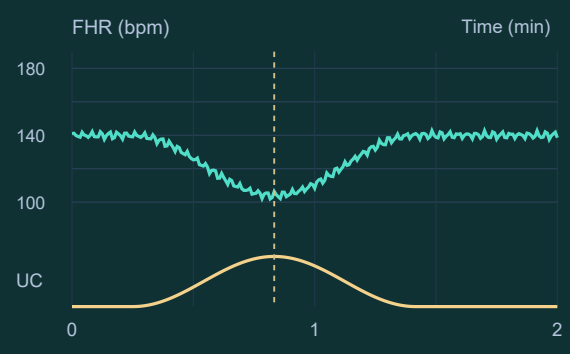


>25 bpm

02 LINE UP THE DIP WITH THE CONTRACTION

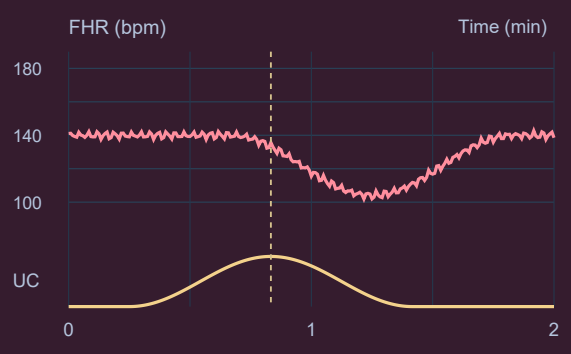
[2, 3]

EARLY



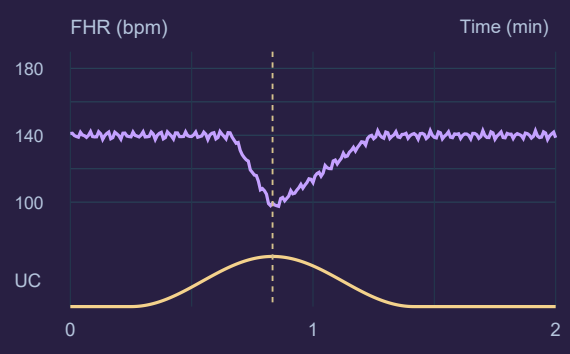
Gradual; mirrors the contraction.
Often **head compression**.

LATE



Gradual; nadir **after** UC peak.
Think uteroplacental insufficiency.

VARIABLE



Abrupt fall; onset to nadir <30 sec.
Often **cord compression**.

Dashed line = contraction peak. Top curve = FHR; lower gold curve = relative uterine activity (UC).

03 NURSING RESPONSE: ASSESS • CORRECT • REASSESS

[1]

CHECK + CALL

Verify fetal signal vs maternal pulse.
Check vitals; alert the obstetric team to concerning patterns.

REPOSITION

Turn laterally; avoid lying flat.

Reassess the fetal response.

PAUSE STIMULATION

Reduce/stop oxytocin for concerning FHR changes or tachysystole, per protocol.

TREAT THE CAUSE

Ordered IV fluid for hypotension; treat fever. Recurrent variables may need ordered amnioinfusion.

OXYGEN

Use for **maternal hypoxia**.

Not routine for II/III when maternal oxygenation is normal.

ESCALATE

If III persists despite corrective care, prepare for **expedited birth**.
Continue monitoring + reassessment.

RECURRENT

Dips with **≥50%** of contractions during a **20-minute** window.

PROLONGED DIP

FHR fall **≥15 bpm** for **2 to <10 min**.
At ≥10 min: baseline change.

TACHYSYSTOLE

More than 5 contractions / 10 min, averaged over **30 min**.

New sustained drop? Respond promptly; do not wait for a category duration threshold. Follow local protocols.

REFERENCES | Click a title in the PDF; full links are in the companion text file.

1. ACOG Clinical Practice Guideline No. 10 (2025): Interpretation and management
2. NICHD / ACOG / SMFM workshop report (Macones et al., 2008)
3. AWHONN Washington: Fetal Monitoring teaching handout (Sperling, 2023)

Prepared October 21, 2026 | Intrapartum study aid. Teaching schematics are not diagnostic patient recordings.